Evan M. Reeves

Quantum Computing Researcher

Experimental hardware • Real-hardware benchmarking • Qubit simulation and control

Hoffman Estates, Illinois • 847.454.5679 • Evan.M.Reeves@gmail.com

Website • Google Scholar • ORCID • LinkedIn

Research Profile

Experimental quantum researcher and licensed Professional Engineer building accessible instruments and evaluating what real quantum processors can deliver. Work spans low-cost NMR and ODMR hardware, multi-architecture benchmarking, superconducting-qubit simulation, and QICK-based control.

Education

Ph.D. in Electrical Engineering, expected 2027 **2023–Present**
University of Illinois Chicago *Chicago, IL*
Advisor: Thomas A. Searles. Research in accessible quantum hardware, multi-architecture benchmarking, and qubit simulation and control.

M.S. in Physics **2015**
Northern Illinois University *DeKalb, IL*
Advisor: Laurence Lurio. Graduate research in materials characterization and particle detection.

B.S. in Applied Physics **2011**
Northern Illinois University *DeKalb, IL*
Minors in Applied Mathematics and Japanese Studies. Undergraduate research in transmission electron microscopy.

Additional education: M.P.S., Purdue Global (2025); Master in Big Data and Business Intelligence, ENEB (2024); Executive M.B.A., Quantic (2023); A.E.S., Harper College (2012).

Research and Entrepreneurship Experience

Research Assistant **2024–Present**
University of Illinois Chicago *Chicago, IL*

- Develop hardware-aware benchmarking workflows for the Quantum Permutation Algorithm across multiple quantum architectures, including execution and performance analysis on IBM superconducting hardware.
- Design and integrate low-cost NMR and ODMR instrumentation for undergraduate quantum information science laboratories.
- Perform electromagnetic simulation of superconducting qubits in Ansys HFSS and develop QICK-based control experiments involving pulse timing and qubit cloaking.
- Collaborate on applied quantum research in chemistry, finance, and edge-based state reconstruction; mentor undergraduates across ECE, computer science, and engineering physics.

Co-Founder and Chief Financial Officer **2025–Present**
SpinQuorum *Chicago, IL*

- Translate low-cost quantum-instrumentation research into a desktop NMR platform for education and workforce development.
- Define technical priorities, resource plans, budgets, and organizational strategy for an early-stage quantum hardware initiative.

Independent Researcher, Low-Cost NMR Quantum Computing **2021–Present**
Hoffman Estates, IL

- Designed custom control electronics for a low-cost single-qubit NMR system and supported prototyping, testing, troubleshooting, and technical communication.

Publications and Proceedings

Blackwell, A. N., **Reeves, E. M.**, Khannejad, A., Strickland, E. M., Metwalli, S. A., Gomez, E., Salonis, C., Chase, Z. A., Glasser, R. T., Kirby, B. T., Lohani, S., & Searles, T. A. (2026). *Benchmarking noisy-intermediate scale hardware with the Quantum Permutation Algorithm: A case study on IBM Sherbrooke*. Quantum 2.0 Conference and Exhibition, paper QTu3A.18, Glasgow, United Kingdom.

Viti, I., **Reeves, E. M.**, Elisma, D., Chandler, J., & Searles, T. A. (2026). *A compact, portable spectrometer for NMR measurements in undergraduate education labs.* Accepted, IEEE Quantum Week 2026, Toronto, Canada.

Viti, I., **Reeves, E. M.**, Elisma, D., Chandler, J., & Searles, T. A. (2026). *A low-cost optically detectable magnetic resonance device.* Accepted, IEEE Quantum Week 2026, Toronto, Canada.

Conference Presentations

Oral Presentations

2026 **Reeves, E. M.**, Viti, I., Elisma, D., Chandler, J., & Searles, T. A. "Low-cost quantum information science laboratory devices." AAPT Summer Meeting, Pasadena, CA.

2026 **Reeves, E. M.**, Viti, I., Elisma, D., Chandler, J., & Searles, T. A. "A low-cost nuclear magnetic resonance quantum computer." Chicago and Illinois Sections of AAPT Spring Meeting, Palatine, IL.

2026 **Reeves, E. M.**, Viti, I., Elisma, D., Chandler, J., & Searles, T. A. "A low-cost nuclear magnetic resonance quantum computer." APS Global Physics Summit, Denver, CO.

2026 **Reeves, E. M.**, Viti, I., & Searles, T. A. "A quantum computer you can build at home." Olive-Harvey Quantum Symposium, Chicago, IL.

2025 Blackwell, A. N., **Reeves, E. M.**, et al. "Hardware-aware performance evaluation of the Quantum Permutation Algorithm." 18th U.S. National Congress on Computational Mechanics, Chicago, IL.

2025 **Reeves, E. M.**, Viti, I., Yekeh, C., & Searles, T. A. "A low-cost nuclear magnetic resonance quantum computer." Northrop Grumman-UIC Quantum Technology Meeting, Chicago, IL.

2025 **Reeves, E. M.**, Viti, I., Yekeh, C., & Searles, T. A. "A low-cost nuclear magnetic resonance quantum computer." IBM HBCU Quantum Center QuERC Annual Conference, Chicago, IL.

2025 **Reeves, E. M.**, Viti, I., & Searles, T. A. "A low-cost nuclear magnetic resonance quantum computer." NSBP-NSHP Joint Conference, San Jose, CA.

2024 **Reeves, E. M.**, Flanigan, A., & Searles, T. A. "Qubit cloaking on superconducting qubits using Qiskit Pulse and Superstaq." NSBP-NSHP Joint Conference, Houston, TX.

Poster Presentations

2026 Regmi, S., **Reeves, E. M.**, Meena, A., Zorzetti, S., Lohani, S., & Searles, T. A. "AI-assisted scalable quantum state reconstruction at edge devices." University of Illinois Superconducting Circuits Conference, Urbana, IL.

2025 **Reeves, E. M.**, Viti, I., & Searles, T. A. "A quantum computer you can build at home." Fermilab SQMS Center Annual Meeting, Batavia, IL.

2025 Blackwell, A. N., **Reeves, E. M.**, et al. "Hardware-aware performance evaluation of the Quantum Permutation Algorithm." Chicago Quantum Symposium, Chicago, IL.

2025 Blackwell, A. N., **Reeves, E. M.**, et al. "Hardware-aware performance evaluation of the Quantum Permutation Algorithm." UIC Computational Research Symposium, Chicago, IL.

2025 Gerding, G., **Reeves, E. M.**, & Searles, T. A. "Qiskit Ansys HFSS finite-element simulation of superconducting qubits." IBM HBCU Quantum Center QuERC Annual Conference, Chicago, IL.

2023 **Reeves, E. M.** "Quantum computing for traffic signal optimization." U.S. Quantum Information Science Summer School, Batavia, IL.

2011 **Reeves, E. M.**, Shurgin, R., & Ito, Y. "Quantitative 3D analysis of nanostructures by TEM anaglyph imaging." NIU Undergraduate Research and Artistry Day, DeKalb, IL.

Contributed and Co-Authored Presentations

2026 Shahbazi, H., Mewali, S., Bagheritabar, M., **Reeves, E. M.**, Prasad, V. K., Searles, T. A., & Chase, Z. A. "Quantum diagonalization-aided design and validation of catalytic pathways for multi-carbon products in CO_2 electroreduction." APS Global Physics Summit, Denver, CO.

2025 Flanigan, A., **Reeves, E. M.**, & Searles, T. A. "Translating two- and three-qubit circuits from Qiskit to Cirq." IBM HBCU Quantum Center QuERC Annual Conference, Chicago, IL.

2025 Yekeh, C., Viti, I., **Reeves, E. M.**, & Searles, T. A. "Development of a low-cost NMR quantum computer." IBM HBCU Quantum Center QuERC Annual Conference, Chicago, IL.

2024 Regmi, S., **Reeves, E. M.**, Zorzetti, S., Lohani, S., & Searles, T. A. "Mitigating multidimensionality challenges of quantum state reconstruction at edge devices." NSBP Annual Conference, Houston, TX.

Engineering and Organizational Experience

Civil Engineer III **2024–Present**
Illinois Department of Transportation *Schaumburg, IL*

- Lead staff, workflows, reporting, and quality-assurance testing for roadway materials supporting infrastructure contracts valued at more than $400 million in an accredited laboratory environment.

Earlier Engineering Roles **2016–2024**
Illinois Department of Transportation *District 1, IL*
Electrical Engineer (2023–2024); Civil Construction Resident Engineer (2021); Senior Engineering Technician (2016–2020)

- Developed systems for inventory, infrastructure mapping, machine-learning aggregate detection, calibration, automation, and laboratory modernization using Python, SQL, Access, Fortran, VB.NET, and VBA.
- Oversaw roadway and bridge field execution and conducted statewide and District 1 research on concrete resistivity, the Illinois Flexibility Index Test, and testing uniformity.

Co-Founder and Chief Executive Officer **2015–2022**
Attic Entertainment, LLC *Hoffman Estates, IL*

- Produced a three-day gaming convention for more than 1,000 attendees; led more than 100 employees and contractors and managed RFP-based site selection, budgets, vendors, and operations.

Graduate and Undergraduate Researcher **2011–2014**
Northern Illinois University *DeKalb, IL*

- Characterized ceramic perovskites using x-ray diffraction and thermogravimetric analysis; supported QuarkNet detector software and produced TEM-based 3D reconstructions of MgO nanostructures.

Teaching and Mentorship

- Mentor undergraduate researchers in computer science, electrical engineering, and engineering physics across multiple institutions, including a 2025 UIC electrical-engineering capstone team.
- Designated IDOT laboratory trainer for testing procedures and annual activities supporting AASHTO accreditation review (2018–2021; 2024–Present).
- Academic tutor in calculus, engineering, physics, statistics, and related subjects for secondary through graduate-level learners (2012–2021).
- Graduate Teaching Assistant for undergraduate electricity and magnetism laboratories at Northern Illinois University (2012).

Licenses, Certifications, Service, and Honors

Licenses: Professional Engineer, Illinois (2025); Professional Engineer, Wyoming (2022).

Certifications: Project Management Professional (2020); Certified Meeting Professional (2021).

Selected service: Union Steward and Bargaining Committee Member, UIC Graduate Employee Organization (2025–2026); Founder and Program Director, The GG Initiative (2020).

Honor: Teamsters Local 916 Scholarship (2025).

Selected affiliations: IEEE; APS; AAPT; ACM; NSBP; NSBE; NSPE; PMI.

Technical Skills

Quantum and scientific computing: Qiskit, QICK, Cirq, PennyLane, Amazon Braket SDK, qBraid, Superstaq, QASM, Ansys HFSS.

Programming and hardware description: Python, SQL, C#, C++, Java, Fortran, R, VBA, VB.NET, Verilog, VHDL, LaTeX, HTML, CSS, XML.

Engineering and analysis software: AutoCAD, FreeCAD, SolidWorks, Maple, Mathcad, Mathematica, Octave, PSpice.

Experimental methods: NMR and ODMR system development, optical microscopy, AFM, SEM, TEM, x-ray diffraction, thermogravimetric analysis.